

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

Video ID:	f3020c65-4c46-40de-8fb8-57135398af55
Total Frames:	2583
Frame Rate:	25.0 fps (assumed)
Duration:	103.3 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1640	✓
RT-DETR Detections	1861	✓
MedSAM2 Detections	2110	✓
Consensus (3-Model Agreement)	4	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	1	25.0%
LOW RISK	3	75.0%
TOTAL ANALYZED	4	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 34 frames)



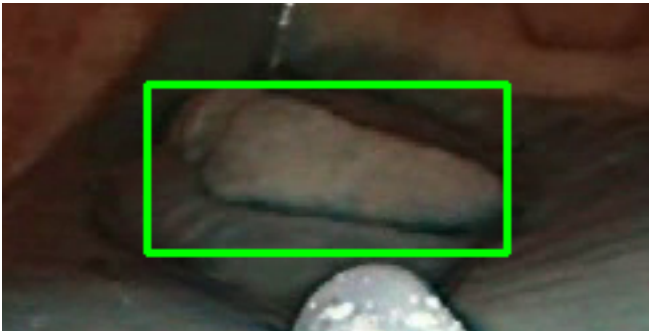
Feature	Value	Interpretation
Frame Index	2009	Frame 2009 of 2583
Detection Method	Detector Consensus (MedSAM2 + U-Net)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 65.3%
Redness Score	0.024	Color-based vascularization
Relative Radius	22.7%	Polyp size as % of frame diagonal
Texture Score	0.323	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	86.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 1019 frames)



Feature	Value	Interpretation
Frame Index	781	Frame 781 of 2583
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 61.4%
Redness Score	0.057	Color-based vascularization
Relative Radius	13.7%	Polyp size as % of frame diagonal
Texture Score	0.041	Surface morphology
Vessel Visibility	0.002	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified

Detection Confidence	85.1%	YOLO / RT-DETR / track confidence
----------------------	-------	-----------------------------------

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 28 frames)

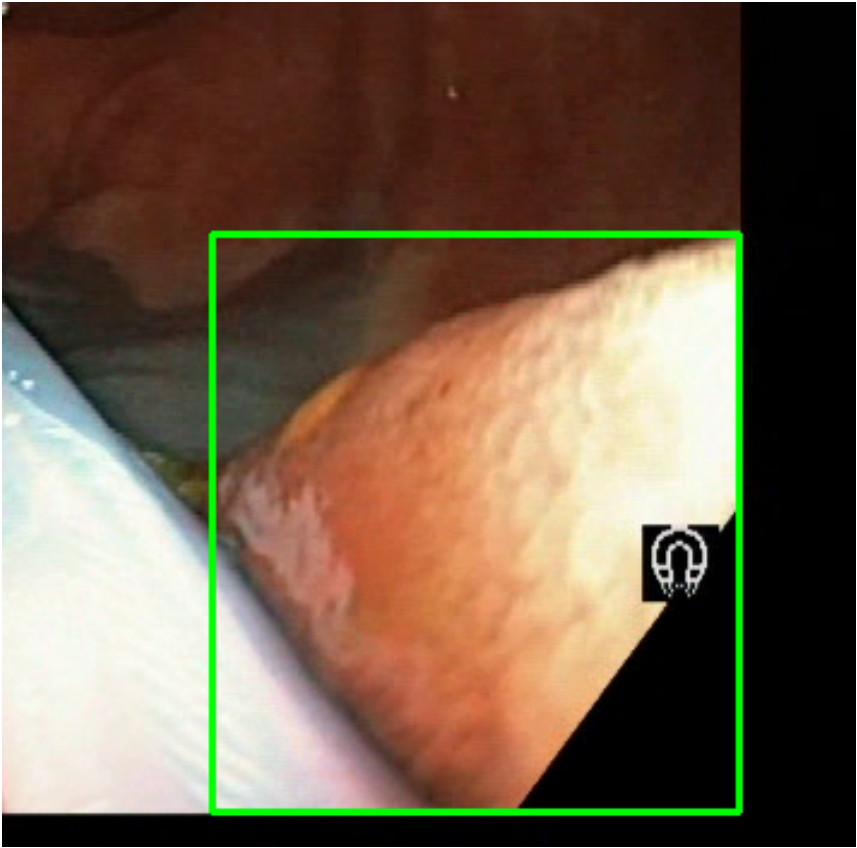


Feature	Value	Interpretation
Frame Index	1062	Frame 1062 of 2583
Detection Method	Detector Consensus (YOLO / RT-DETR)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 62.9%
Redness Score	0.056	Color-based vascularization
Relative Radius	6.0%	Polyp size as % of frame diagonal
Texture Score	0.106	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	90.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 55 frames)



Feature	Value	Interpretation
Frame Index	1132	Frame 1132 of 2583
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	SERRATED_POLYP	Type confidence: 92.0%
Redness Score	0.162	Color-based vascularization
Relative Radius	38.4%	Polyp size as % of frame diagonal
Texture Score	0.354	Surface morphology
Vessel Visibility	0.454	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output

Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	82.6%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 4 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 1 (25.0%)
- LOW RISK polyps: 3 (75.0%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.